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A THREE-DIMENSIONAL KINEMATIC METHOD FOR DETERMINING THE EFFECTIVENESS OF ARM SEGMENT ROTATIONS IN PRODUCING RACQUET-HEAD SPEED

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Abstract—The contribution that a segment's anatomical rotations make to racquet-head speed depends on both the segment's angular velocity and the instantaneous position of the head of the racquet with respect to the segment's axes of rotation. Any analysis of racquet swing technique that does not consider both of these factors simultaneously is, at best, incomplete. With this in mind, a three-dimensional kinematic method was developed to determine the effectiveness of the anatomical rotations of the upper arm, forearm, and hand in producing racquet-head speed. The method entailed developing a system of vector equations for three-dimensional upper limb rotations that used displacement histories of 10 selected landmarks as input. The required three-dimensional displacement histories were obtained using three cine cameras and the DLT approach. To test the diagnostic capabilities of the method, a tennis serve was selected for analysis. For the player and serve analyzed, the greatest contribution to racquet-head speed at impact was produced by internal rotation of the upper arm (8 m s^{-1}). Forearm pronation, although exhibiting the fastest rotation at impact (24 rad s^{-1}), ranked only fourth in terms of its contribution (4 m s^{-1}) to racquet-head speed. To test the performance of the method, a comparison was made between the racquet-head speed measured directly from film and the racquet-head speed computed by summing all of the individual segment contributions to speed commencing at the start of forward swing and ending at ball contact. The results indicate that the method can successfully determine the individual contributions that the different anatomical rotational velocities of the arm segments make to the measured instantaneous racquet-head speed.

NOMENCLATURE

- $\hat{\mathbf{p}}_{ij}$ unit vector of segment i aligned with the anatomical rotation j, where $i=1$(upper arm), 2 (forearm), 3 (hand) j=FE (flexion/extension), AA (adduction/abduction), L (longitudinal), UR (ulnar/radial deviation), VV (varus/valgûs)
- $\boldsymbol{\omega}_i$ absolute angular velocity of segment i
- $\boldsymbol{\omega}_{ij}$ component of absolute angular velocity of segment i aligned with the unit vector representing anatomical rotation j
- $\boldsymbol{\omega}_{ia/ib}$ relative angular velocity between segments ia and ib expressed in the cameras' coordinate system
- $\mathbf{A}\boldsymbol{\omega}_{ij}$ anatomical rotation j of segment i
- $\mathbf{r}_{p/q}$ displacement vector between reference points p and q
- $\mathbf{V}_p$ absolute linear velocity of reference point p

INTRODUCTION

The equations of impact for a semi-elastic collision (Daish, 1972) reveal that the velocity of departure, for an object struck by an implement, is dependent on (a) the relative approach velocities of the two impacting bodies, (b) the masses of the two bodies, and (c) the coefficient of restitution. In racquet sports, the velocity of the incoming ball is a parameter beyond the player's control. The coefficient of restitution (e) of the racquet/ball (or racquet/shuttlecock) combination is dependent on the relative speed and location of impact, as well as the physical properties (stiffness and tension) of the racquet and string. However, for a player of reasonable ability, hitting the ball in the same location on a racquet is a reasonable expectation and thus the value of 'e' will mainly be affected by the relative speed of impact. The speed of the racquet-head at impact is a variable that can be varied considerably by the player through his arm-swing mechanics. As such, the arm rotation pattern used in generating maximum racquet-head speed at impact is of considerable interest to both players and their coaches.

The contributions that different body segments make to the overarm throwing motion have been examined in various ways. Hoshikawa and Toyoshima (1976) and Toyoshima *et al.* (1974) examined the effect that restriction of movement of selected body segments had on throwing velocity. Their method, although appealing in its simplicity, confounds the results with the negative effect that such cumbersome restrictions impose on the normal movement patterns of the other segments. Feltner and Dapena (1986, 1989) used three-dimensional kinetic analyses to better understand the source of the joint torques exerted on the arm segments during the throwing motion in baseball. In their work, the angular velocities about the longitudinal axis of each segment were assumed to be zero. Such an assumption is

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clearly not justifiable for analysis of racquet sports. In tennis, Plagenhoef (1971) analyzed the relative contributions of body segments to the tennis serve by mathematically zeroing the angular velocities and accelerations of each of the body segments in turn. This analysis was based on a two-dimensional representation of the movement sequence. Other researchers (Gowitzke and Waddell, 1979; Jack *et al.*, 1979; van Gheluwe *et al.*, 1987; Woo and Chapman, 1991) have documented the arm segment rotations as well as the associated timing patterns during a typical arm-swing in racquet-ball, badminton, tennis, and squash, respectively. Elliott *et al.* (1986) examined the linear velocities of the hip, elbow, wrist, and racquet during the tennis serve and reported a progressive proximal to distal increase in velocities to impact. However, such results reveal nothing of the individual contributions of segment rotations to racquet-head speed.

The racquet-head speed displayed at impact results from a series of upper limb segment rotations generated by muscle torques. The segment rotations frequently overlap one another, making it difficult to estimate their individual contributions to racquet-head speed. The effectiveness of these rotations in generating racquet-head speed depends both on their magnitude and on the positioning of the racquet-head with respect to the segment's axis of rotation. This last point is important for it means that a kinetic study of the generated joint torques, even assuming they could be computed accurately (Williams and Cavanagh, 1983), would in itself not answer the question as to the effectiveness of the swing mechanics. One might generate substantial torques about a segment's axis, but the effect would be minimal if the racquet-head is close to the axis of rotation. The intent of this paper is to illustrate, by example, a kinematic method for determining the anatomical angular velocities of the racquet-arm segments and the instantaneous contribution that each of these rotations makes to racquet-head speed.

THEORETICAL DEVELOPMENT

The tennis serve has been picked as an example to illustrate the methodology; however, any of the arm-swing patterns from the racquet sports could have been used. The procedure requires a method of measuring the angular velocities of the segments with respect to their anatomical axes. Anatomical axes were chosen for this study as they reflect the joint rotations that a player has control over through muscle actions. As well, the associated anatomical rotations (e.g. flexion/extension, pronation/supination, etc.) are terms that have traditionally been employed by the coaching profession to describe the joint movement patterns of their sport.

Figure 1 depicts the three segments of the right upper limb with a racquet being gripped by the hand segment. Ten reference markers (H, I, E, F, B, A, N, M, L, J), representing landmarks on the arm/racquet system, are identified for future reference in the algorithm. Points G, D, C, O, and K are the computed mid-points between their associated surface landmarks. The procedure for computing the anatomical rotational velocities consisted of first constructing a separate set of orthogonal unit vectors for each of the three segments of the upper limb. The directions of these unit vectors were constructed so as to approximate the different anatomical axes of rotation for each segment. The absolute angular velocities about the segment's axes were computed and used to determine the relative rotations (i.e. anatomical rotations) of the individual segments. The vector cross-products between the anatomical rotational velocities and their respective displacement vectors to the center of the racquet-head were then used to determine the effectiveness of these rotations in producing racquet-head speed.

Upper arm

An orthogonal unit vector axis system for the upper arm was constructed with its origin at the gleno-

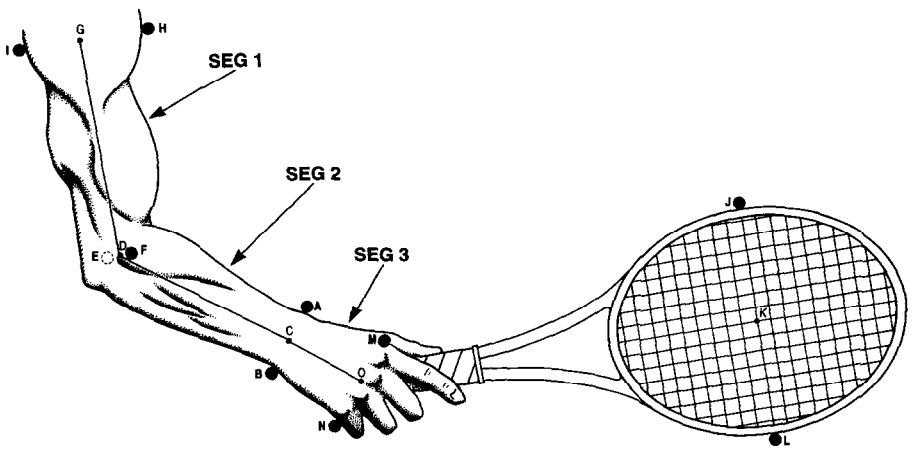

Fig. 1. Location of the markers on the upper limb and racquet of the tennis player.

humeral joint. The first unit vector ($\hat{\mathbf{p}}_{1L}$), along the longitudinal axis, was determined by

$$\hat{\mathbf{p}}_{1L}=\frac{\mathbf{r}_{D/G}}{|\mathbf{r}_{D/G}|}.$$

A second unit vector, $\hat{\mathbf{p}}_{1AA}$, was constructed perpendicular to $\hat{\mathbf{p}}_{1L}$ and, for the classic anatomical position, represents the direction of the rotation axis for upper arm adduction/abduction:

$$\hat{\mathbf{p}}_{1AA}=\frac{\hat{\mathbf{p}}_{1L}\times\mathbf{r}_{E/D}}{|\hat{\mathbf{p}}_{1L}\times\mathbf{r}_{E/D}|}.$$

A third unit vector, orthogonal to $\hat{\mathbf{p}}_{1L}$, and $\hat{\mathbf{p}}_{1AA}$, was likewise constructed and, for the classic anatomical position, represents the direction of the rotation axis for upper arm flexion/extension:

$$\hat{\mathbf{p}}_{1FE}=\frac{\hat{\mathbf{p}}_{1AA}\times\hat{\mathbf{p}}_{1L}}{|\hat{\mathbf{p}}_{1AA}\times\hat{\mathbf{p}}_{1L}|}.$$

Since the axis system for the upper arm, as defined by these unit vectors, is fixed directly to the segment itself, the anatomical description for the measured orthogonal rotations associated with $\hat{\mathbf{p}}_{1AA}$ and $\hat{\mathbf{p}}_{1FE}$ must be determined from a visual record of the performance. For example, by noting the direction of the axis EF attached to the distal end of the humerus one can assign an anatomical description to the motion that most closely fits the rotation of the upper arm about a parallel axis through the shoulder joint. Consideration was given to attaching the upper arm's reference axis to the shoulder complex, but the idea was rejected because of the shoulder's lack of rigidity. In practice, the limitation of requiring a visual record of the upper arm movement does not pose any real difficulty since film or video will be required to measure displacement and velocity vectors of the reference landmarks.

Next, the algorithm requires the relative velocity (three-dimensional) of reference point E with respect to D ($\mathbf{V}_{E/D}$) and is acquired externally from either film or video. Further details on this are presented in the Experimental Procedure section of the paper. The portion of $\mathbf{V}_{E/D}$ tangential to $\mathbf{r}_{E/D}$ is computed ($V_{t(E/D)}=\mathbf{V}_{E/D}\cdot\hat{\mathbf{p}}_{1AA}$) and used to solve for the magnitude of the longitudinal angular velocity of the upper arm (ω_{1L}).

$$\omega_{1L}=\frac{V_{t(E/D)}}{|\mathbf{r}_{E/D}|}.$$

The vector representing this rotation is

$$\boldsymbol{\omega}_{1L}=(\omega_{1L})*(\hat{\mathbf{p}}_{1L}). \quad (1)$$

The other two orthogonal components of the upper arm angular velocity were computed as follows:

Let $\boldsymbol{\omega}_1$ be the total absolute angular velocity of the upper arm

$$\boldsymbol{\omega}_1=\omega_{1AA}*(\hat{\mathbf{p}}_{1AA})+\omega_{1FE}*(\hat{\mathbf{p}}_{1FE})+\omega_{1L}*(\hat{\mathbf{p}}_{1L}). \quad (2)$$

Using the identity $\boldsymbol{\omega}_1\times\mathbf{r}_{D/G}=\mathbf{V}_{D/G}$:

$$\omega_{1AA}*(\hat{\mathbf{p}}_{1AA}\times\mathbf{r}_{D/G})+\omega_{1FE}*(\hat{\mathbf{p}}_{1FE}\times\mathbf{r}_{D/G})+\omega_{1L}*(\hat{\mathbf{p}}_{1L}\times\mathbf{r}_{DG})=\mathbf{V}_{D/G}. \quad (3)$$

By taking the scalar product of equation (3) with the unit vector $\hat{\mathbf{p}}_{1AA}$, and noting that $\hat{\mathbf{p}}_{1L}$ is parallel to $\mathbf{r}_{D/G}$, an expression for '$\boldsymbol{\omega}_{1FE}$' is obtained:

$$\boldsymbol{\omega}_{1FE}=\left[\frac{\hat{\mathbf{p}}_{1AA}\cdot\mathbf{V}_{D/G}}{\hat{\mathbf{p}}_{1AA}\cdot(\hat{\mathbf{p}}_{1FE}\times\mathbf{r}_{D/G})}\right]*[\hat{\mathbf{p}}_{1FE}]. \quad (4)$$

Similarly, taking the scalar product of equation (3) with $\hat{\mathbf{p}}_{1FE}$ results in an expression for '$\boldsymbol{\omega}_{1AA}$':

$$\boldsymbol{\omega}_{1AA}=\left[\frac{\hat{\mathbf{p}}_{1AA}\cdot\mathbf{V}_{D/G}}{\hat{\mathbf{p}}_{1FE}\cdot(\hat{\mathbf{p}}_{1AA}\times\mathbf{r}_{D/G})}\right]*[\hat{\mathbf{p}}_{1AA}]. \quad (5)$$

The direction of the anatomical rotations associated with the vectors $\boldsymbol{\omega}_{1L}, \boldsymbol{\omega}_{1AA}$, and $\boldsymbol{\omega}_{1FE}$ can be determined by examining the sign of the dot product between $\boldsymbol{\omega}_1$ and the appropriate unit vector ($\hat{\mathbf{p}}_{1L}, \hat{\mathbf{p}}_{1AA}$, or $\hat{\mathbf{p}}_{1FE}$).

Forearm

The reference axis system for the forearm segment, consisting of orthogonal unit vectors $\hat{\mathbf{p}}_{2L}$, $\hat{\mathbf{p}}_{2VV}$, and $\hat{\mathbf{p}}_{2FE}$, has its origin at the humeral–ulnar joint. Unit vector $\hat{\mathbf{p}}_{2FE}$ is fixed to the distal end of the humerus since its construction can be traced back to its dependence on $\mathbf{r}_{E/D}$. Unit vector $\hat{\mathbf{p}}_{2L}$ is fixed to the forearm segment, and $\hat{\mathbf{p}}_{2VV}$ is orthogonal to these two:

$$\hat{\mathbf{p}}_{2L}=\frac{\mathbf{r}_{C/D}}{|\mathbf{r}_{C/D}|},$$

$$\hat{\mathbf{p}}_{2VV}=\frac{\hat{\mathbf{p}}_{2L}\times\hat{\mathbf{p}}_{1FE}}{|\hat{\mathbf{p}}_{2L}\times\hat{\mathbf{p}}_{1FE}|},$$

$$\hat{\mathbf{p}}_{2FE}=\frac{\hat{\mathbf{p}}_{2VV}\times\hat{\mathbf{p}}_{2L}}{|\hat{\mathbf{p}}_{2VV}\times\hat{\mathbf{p}}_{2L}|}.$$

Similar equations to those developed previously for the upper arm rotations are shown below for the forearm segment.

Longitudinal rotation:

$$\boldsymbol{\omega}_{2L}=\omega_{2L}*(\hat{\mathbf{p}}_{2L}). \quad (6)$$

Valgus/varus:

It was assumed that valgus/varus rotation at the elbow joint would be negligible.

$$\boldsymbol{\omega}_{2VV}=0. \quad (7)$$

Flexion/extension:

$$\boldsymbol{\omega}_{2FE}=\left[\frac{\hat{\mathbf{p}}_{2VV}\cdot\mathbf{V}_{C/D}}{\hat{\mathbf{p}}_{2VV}\cdot(\hat{\mathbf{p}}_{2FE}\times\mathbf{r}_{C/D})}\right]*[\hat{\mathbf{p}}_{2FE}]. \quad (8)$$

Hand

The anatomical rotations of the hand consist only of flexion/extension and ulnar/radial deviation (Norkin and Levangie, 1992). These rotations are defined in

anatomical textbooks only with respect to displacement away from the neutral configuration of the wrist joint. As such, an axis system specifying the hand segment's rotation could either be fixed in the hand or in the distal end of the forearm since both axes systems would measure the same result when rotation commences from the neutral configuration. However, when a wrist joint rotation commences from a non-neutral configuration, there is no commonly accepted axes convention that describes the orientation of the functional (i.e. anatomical) axes (Andrews, 1983; Grood and Suntay, 1983). For the purposes of this paper, a decision was made to fix the axis system, representing rotation of the hand, to the distal end of the forearm. It was felt that practitioners (i.e. coaches and athletes) interpret flexion/extension and ulnar/radial deviation as motion of the hand with respect to the styloid processes of the ulna and radius. Such an axis system represents ulnar/radial deviation as a projection of the angular displacement of the longitudinal axis of the hand onto the plane corresponding to ulnar/radial deviation from the neutral position. As anatomical rotations about the wrist are only defined with respect to the neutral configuration, this approach would seem reasonable—especially for a functional rotation commonly referred to as ulnar/radial deviation. Similar reasoning can be presented for the axis representing flexion/extension of the wrist.

The unit vectors for this axis system are:

$\hat{\mathbf{p}}_{3L}$, which was set equal to $\hat{\mathbf{p}}_{2L}$, since the hand segment is incapable of initiating its own longitudinal rotation.

$\hat{\mathbf{p}}_{3FE}$ first approximated by

$$\hat{\mathbf{p}}_{3FE}=\frac{\mathbf{r}_{B/A}}{|\mathbf{r}_{B/A}|},$$

$\hat{\mathbf{p}}_{3UR}$, computed by

$$\hat{\mathbf{p}}_{3UR}=\frac{\hat{\mathbf{p}}_{2L}\times\hat{\mathbf{p}}_{3FE}}{|\hat{\mathbf{p}}_{2L}\times\hat{\mathbf{p}}_{3FE}|}$$

and then $\hat{\mathbf{p}}_{3FE}$, in its final orthogonal form

$$\hat{\mathbf{p}}_{3FE}=\frac{\hat{\mathbf{p}}_{3UR}\times\hat{\mathbf{p}}_{2L}}{|\hat{\mathbf{p}}_{3UR}\times\hat{\mathbf{p}}_{2L}|}.$$

Equations similar to those developed previously for the upper arm and forearm rotations are shown below for the hand segment.

Longitudinal:

$$\boldsymbol{\omega}_{3L}=\boldsymbol{\omega}_{2L}. \tag{9}$$

Flexion/extension:

$$\boldsymbol{\omega}_{3FE}=\left[\frac{\hat{\mathbf{p}}_{3UR}\cdot\mathbf{V}_{O/C}-\hat{\mathbf{p}}_{3UR}\cdot(\boldsymbol{\omega}_{2L}\times\mathbf{r}_{O/C})}{\hat{\mathbf{p}}_{3UR}\cdot(\hat{\mathbf{p}}_{3FE}\times\mathbf{r}_{O/C})}\right]*[\hat{\mathbf{p}}_{3FE}]. \tag{10}$$

Ulnar/radial deviation:

$$\boldsymbol{\omega}_{3UR}=\left[\frac{\hat{\mathbf{p}}_{3FE}\cdot\mathbf{V}_{O/C}-\hat{\mathbf{p}}_{3FE}\cdot(\boldsymbol{\omega}_{2L}\times\mathbf{r}_{O/C})}{\hat{\mathbf{p}}_{3FE}\cdot(\hat{\mathbf{p}}_{3UR}\times\mathbf{r}_{O/C})}\right]*[\hat{\mathbf{p}}_{3UR}]. \tag{11}$$

Having determined the angular velocities of the segments, their respective contributions to the development of racquet-head speed can be determined by the following process.

If we assume that the racquet is being held with a firm grip such that no rotation occurs between the racquet handle and the hand, then the velocity of the middle of the racquet head (point K) can be expressed as

$$\mathbf{V}_K=\mathbf{V}_G+\boldsymbol{\omega}_1\times\mathbf{r}_{D/G}+\boldsymbol{\omega}_2\times\mathbf{r}_{C/D}+\boldsymbol{\omega}_3\times\mathbf{r}_{K/C}, \tag{12}$$

where $\boldsymbol{\omega}_1$, $\boldsymbol{\omega}_2$, and $\boldsymbol{\omega}_3$ are absolute angular velocities of the three segments expressed in the reference system units of the external measurement system. In most cases, this would be the reference coordinates of the three-dimensional camera (cine or video) system.

From the known relationship:

$$\boldsymbol{\omega}_2=\boldsymbol{\omega}_1+\boldsymbol{\omega}_{2/1}$$

and

$$\boldsymbol{\omega}_3=\boldsymbol{\omega}_2+\boldsymbol{\omega}_{3/2},$$

$$\boldsymbol{\omega}_3=\boldsymbol{\omega}_1+\boldsymbol{\omega}_{2/1}+\boldsymbol{\omega}_{3/2}.$$

Substituting into equation (12), and noting that

$$\mathbf{r}_{K/G}=\mathbf{r}_{D/G}+\mathbf{r}_{C/D}+\mathbf{r}_{K/C}$$

and

$$\mathbf{r}_{K/D}=\mathbf{r}_{C/D}+\mathbf{r}_{K/C},$$

one obtains the fundamental equation describing the relationship between the individual segment rotations and the linear velocity of the racquet head:

$$\mathbf{V}_K=\mathbf{V}_G+\boldsymbol{\omega}_1\times\mathbf{r}_{K/G}+\boldsymbol{\omega}_{2/1}\times\mathbf{r}_{K/D}+\boldsymbol{\omega}_{3/2}\times\mathbf{r}_{K/C}, \tag{13}$$

where $\mathbf{V}_G$ is interpreted as the contribution that the legs and torso make to developing racquet-head velocity. $\boldsymbol{\omega}_{2/1}$ and $\boldsymbol{\omega}_{3/2}$ are relative angular velocity terms reflecting rotations of the forearm and hand segments, respectively. Their values, as well as that of $\boldsymbol{\omega}_1$, can be determined as follows:

$$\boldsymbol{\omega}_1=\boldsymbol{\omega}_{1L}+\boldsymbol{\omega}_{1FE}+\boldsymbol{\omega}_{1AA},$$

$$\boldsymbol{\omega}_{2/1}=\boldsymbol{\omega}_2-\boldsymbol{\omega}_1,$$

$$\boldsymbol{\omega}_{2/1}=(\boldsymbol{\omega}_{2L}+\boldsymbol{\omega}_{2FE}+\boldsymbol{\omega}_{2VV})-(\boldsymbol{\omega}_{1L}+\boldsymbol{\omega}_{1FE}+\boldsymbol{\omega}_{1AA}),$$

$$\boldsymbol{\omega}_{3/2}=\boldsymbol{\omega}_3-\boldsymbol{\omega}_2,$$

$$\boldsymbol{\omega}_{3/2}=(\boldsymbol{\omega}_{2L}+\boldsymbol{\omega}_{3FE}+\boldsymbol{\omega}_{3UR})-(\boldsymbol{\omega}_{2L}+\boldsymbol{\omega}_{2FE}+\boldsymbol{\omega}_{2VV}).$$

This last step eliminates extraneous rotations produced by the muscle groups of another joint. For example, internal rotation of the upper arm will also produce rotations of the forearm and hand. In the above algorithm, the associated musculature of the forearm and hand will not receive credit for rotations they did not produce.

The relative angular velocity values ($\boldsymbol{\omega}_{2/1}$ and $\boldsymbol{\omega}_{3/2}$) are expressed in the absolute coordinate system of the external measuring system (i.e. the camera's). The following transforms them in their anatomical coordinate system.

Upper arm

$\boldsymbol{\omega}_1$ is an absolute term and already partitioned into anatomical components ($\boldsymbol{\omega}_{1L}$, $\boldsymbol{\omega}_{1FE}$, and $\boldsymbol{\omega}_{1AA}$).

Forearm

(i) Pronation/supination:

$$\mathbf{A}\boldsymbol{\omega}_{2L} = (\boldsymbol{\omega}_{2/1} \cdot \hat{\mathbf{p}}_{2L}) * \hat{\mathbf{p}}_{2L}.$$

(ii) flexion/extension of the forearm

$$\mathbf{A}\boldsymbol{\omega}_{2FE} = (\boldsymbol{\omega}_{2/1} \cdot \hat{\mathbf{p}}_{2FE}) * \hat{\mathbf{p}}_{2FE}.$$

(iii) valgus/varus rotations are assumed to be zero

$$\mathbf{A}\boldsymbol{\omega}_{2VV} = 0.$$

Thus,

$$\boldsymbol{\omega}_{2/1} = \mathbf{A}\boldsymbol{\omega}_{2L} + \mathbf{A}\boldsymbol{\omega}_{2FE}.$$

Hand

(i) Flexion/extension:

$$\mathbf{A}\boldsymbol{\omega}_{3FE} = (\boldsymbol{\omega}_{3/2} \cdot \hat{\mathbf{p}}_{3FE}) * \hat{\mathbf{p}}_{2FE}.$$

(ii) Ulnar/radial deviation

$$\mathbf{A}\boldsymbol{\omega}_{3UR} = (\boldsymbol{\omega}_{3/2} \cdot \hat{\mathbf{p}}_{3UR}) * \hat{\mathbf{p}}_{3UR}.$$

(iii) Longitudinal rotations are assumed to be zero

$$\mathbf{A}\boldsymbol{\omega}_{3L} = 0.$$

Thus,

$$\boldsymbol{\omega}_{3/2} = \mathbf{A}\boldsymbol{\omega}_{3FE} + \mathbf{A}\boldsymbol{\omega}_{3UR}.$$

The contribution that each of these separate anatomical rotational components ($\mathbf{A}\boldsymbol{\omega}$) make to racquet-head velocity can be determined by substituting into equation (13):

$$\begin{aligned}\mathbf{V}_K = \mathbf{V}_G &+ \{(\boldsymbol{\omega}_{1L} + \boldsymbol{\omega}_{1FE} + \boldsymbol{\omega}_{1AA}) \times \mathbf{r}_{K/G}\} \\ &+ \{(\mathbf{A}\boldsymbol{\omega}_{2L} + \mathbf{A}\boldsymbol{\omega}_{2FE}) \times \mathbf{r}_{K/D}\} \\ &+ \{(\mathbf{A}\boldsymbol{\omega}_{3FE} + \mathbf{A}\boldsymbol{\omega}_{3UR}) \times \mathbf{r}_{K/C}\}. \qquad (14)\end{aligned}$$

The algorithm developed in this paper relies upon the following assumptions: (a) the constructed orthogonal axes for the upper limb segments closely approximate their anatomical axes; (b) the valgus/varus rotation at the elbow joint is zero; (c) the longitudinal rotation of the hand about the wrist joint is zero; (d) the hand and the racquet are a single rigid body.

EXPERIMENTAL PROCEDURE

The flat, first service of an Australian 'A'-grade tennis player, whose serve was known *a priori* to lack speed, was filmed using three phase-locked 16 mm cameras operating at a nominal speed of 200 fps. As well, test sequences of isolated rotations of the upper limb segments (e.g. pronation/supination, flexion/extension, etc.), performed at approximately 0.5 rad s^{-1} about a single axis, were filmed to serve as a check on the computational procedure. The cameras used 12 factor shutters and were positioned on the racquet–arm side in front, beside, and behind the subject at different levels (2.0, 4.0, 1.5 m) above the floor. Eight reflective hemispheres, 17 mm in diameter, were glued to the subjects skin (Fig. 1, points H, I, E, F, B, A, N, M). Two other markers (points L, J) were fastened to the sides of the head of the racquet. The three-dimensional coordinate values for these landmarks were determined using the direct linear transformation method (Abdel-Aziz and Karara, 1971) and a large calibration frame previously described (Wood and Marshall, 1986). The average standard error of the three-dimensional reconstruction process (30 control points), as determined from the calibration frame, was ± 4 mm. This reconstruction error value is of similar magnitude as that previously reported by Shapiro (1978) and Miller *et al.* (1980). The calibration frame was elevated 0.9 m above the floor so that the subject's racquet–arm system would remain within the calibrated volume during the forward swing sequence. The forward swing sequence, commencing at first forward movement by the arm and finishing approximately 5 frames past ball contact, was digitized manually from a projected image 1/9 that of life size. Coordinate values for each surface landmark were recorded for each frame of film from all three cameras. For a marker momentarily hidden from view from a camera, an educated guess to its location was used. The data points were filtered using the optimally regularized quintic spline subroutine GCVSPL (Woltring, 1986).

RESULTS

The results from the isolated anatomical test rotations, which had been performed slowly and in the best view of the three cameras, indicated that the mathematical algorithm was performing correctly. Under these controlled conditions, the only detected angular velocity was about the segment axis selected for testing. These preliminary tests were a necessary check but in themselves not capable of revealing the magnitude of measurement error in the tennis serve itself. The tennis serve is characterized by much greater segment rotation rates and has occurrences of surface markers hidden from camera views. To obtain an indication of the error in the rotational measurements of the tennis serve, the hypothetical varus/valgus rotation of the forearm was computed over the duration of the serve. In practice, this rotation should be zero; thus any computed angular velocity about this axis is an indication of the magnitude of error associated with the procedure. The orthogonal components of the RMS component values for the valgus/varus rotational velocity computed over the 62 frames of film were 1.97, 1.85, and 0.90 rad s^{-1}, respectively. Most of this error can be attributed to the momentary losses from sight of the medial elbow marker for one or more of the three cameras during the forward arm-swing.

As mentioned in the Methods section, the anatomical description for the measured upper arm segment rotations had to be determined from the film record (Fig. 2). Upon inspection, it was evident that the three orthogonal rotational components $A\omega_{1FE}$, $A\omega_{1AA}$, and $A\omega_{1L}$ were primarily shoulder flexion/extension (flex/ext), horizontal adduction/abduction (Hadd/Habd), and internal/external rotation (inter/exter), respectively.

The computed anatomical rotational velocities for the upper arm, forearm, and hand during the forward swing revealed that the highest overall rotational velocities were obtained by the hand segment during periods of ulnar deviation (40 rad s^{-1}) and flexion (25 rad s^{-1}) of the wrist joint (Fig. 3). However, at the moment of impact the highest anatomical rotational velocities (Fig. 3) were recorded for forearm pronation (24 rad s^{-1}), upper arm internal rotation (22 rad s^{-1}),

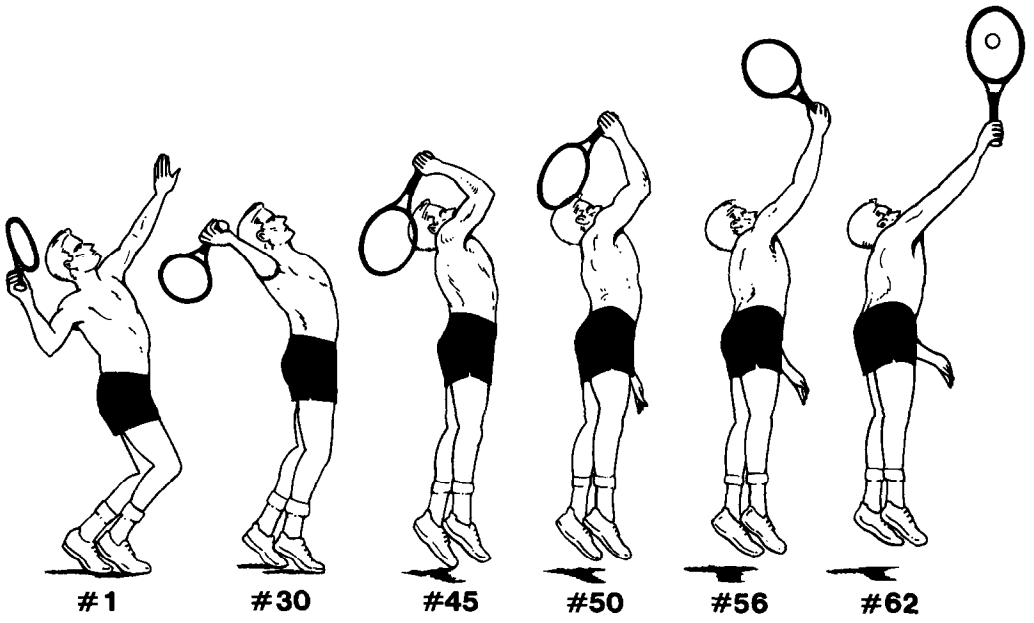

Fig. 2. Sequential images of the subject's tennis serve commencing at the start of the forward arm-swing and ending at ball contact.

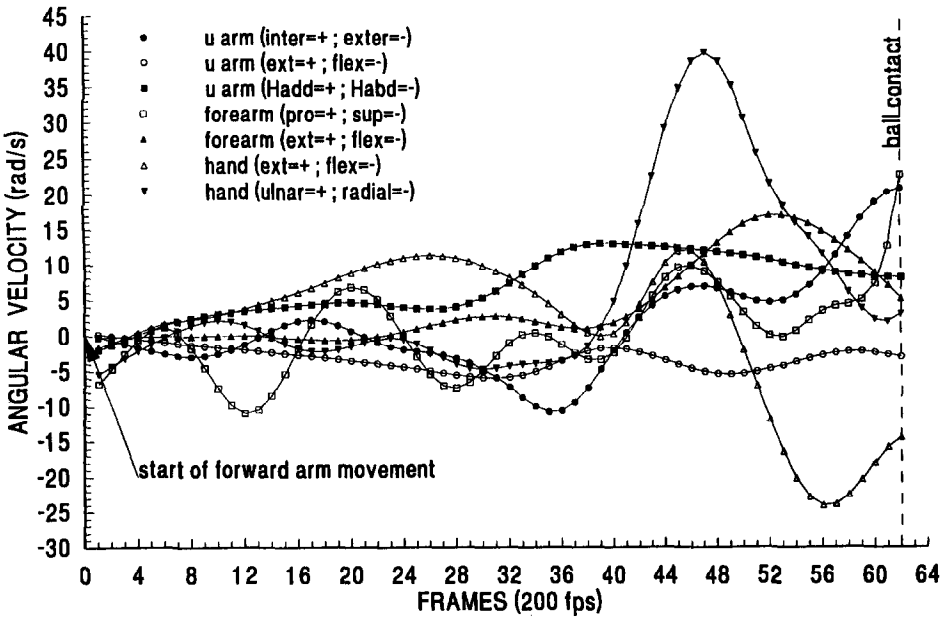

Fig. 3. Rotational velocities of the upper arm, forearm, and hand about their respective anatomical axes during the forward upper limb movement of the tennis serve. Time interval between frames is 1/200 s.

hand flexion (15 rad s^{-1}), and upper arm horizontal adduction (9 rad s^{-1}).

For this particular tennis player, the greatest contributor to the racquet-head's forward speed at ball contact (Fig. 4) was upper-arm internal rotation (8 m s^{-1}), followed in turn by wrist flexion (7 m s^{-1}), upper-arm horizontal adduction (6.5 m s^{-1}), forearm pronation (4 m s^{-1}), and shoulder forward movement (2.5 m s^{-1}). The contributions of trunk rotation and flexion/extension of the spine, along with the forward drive from the legs, are not separated out by the present algorithm. However, their net kinematic effect is accounted for by the term $\mathbf{V}_g$ which is the velocity of the right shoulder joint. For this player, shoulder flexion/extension, elbow extension, and wrist ulnar deviation were not direct contributors at impact. One should not conclude that this is the ideal contribution profile for a tennis serve. In fact, the comparison of this player's service profile with that of a world calibre player would most likely point to the lack of contribution from ulnar deviation as a factor that should be addressed in order to improve the speed of his serve.

The computed contributions that each of the anatomical rotations made to racquet-head velocity (Fig. 4) clearly reveal that knowledge of a segment's angular velocity alone is insufficient information for assessing the resulting benefits to racquet-head speed. A comparison of the contributions of pronation of the forearm and horizontal cross-flexion of the upper arm to racquet-head speed (Figs 3 and 4—top and middle), illustrates this point. At the time of ball contact, the forearm is being pronated at approximately 24 rad s^{-1} with a corresponding contribution to racquet-head speed of 4 m s^{-1}. On the other hand, horizontal cross-flexion of the upper arm at 1/3 the speed of the forearm pronation produces a contribution to racquet-head speed of 6.5 m s^{-1}.

A first impression from examining the velocity contribution graphs (Fig. 4) is that a number of anomalies appear to be present in the results. For example, between frames 4–48 (Fig. 4—top), there is a negative contribution to forward racquet-head speed attributed to horizontal adduction of the upper arm. However, the film revealed that during this phase of the swing, the racquet-head was positioned over the left side of his torso and was actually being rotated backward by the horizontal adduction of his right upper arm. It is observed that, for this player, forearm extension at contact was counterproductive to racquet-head speed (Fig. 4—middle). Although this seemingly suggests an error in the algorithm, the film revealed that as the racquet-head approached ball contact, the subject's upper arm had been internally rotated to the extent that the direction of elbow extension produced a lateral and backwards movement of the racquet-head. It is further observed that for this tennis player, the ulnar deviation rotation was counterproductive to racquet-head speed over the last 22 frames before ball contact (Fig. 4—bottom). Between frames 40–52, the subject's racquet-head was positioned well below the hand in what coaches refer to as the 'back-scratch' position. Employing an ulnar deviation movement at this stage rotates the racquet-head backwards. Between frame 53 and ball contact, the strong pronation of the forearm has rotated the wrist joint to a position where ulnar deviation moved the racquet-head laterally and slightly backwards. The lack of rotational contributions of elbow extension and ulnar deviation to forward racquet-head speed would most certainly appear to be technical flaws in this player's technique. A comparison of the profile taken from a world class server could possibly suggest at what stages of the swing these two segment rotations should be altered so as to produce greater racquet-head speed at contact. The identification of swing anomalies such as this points to the potential strength of this analytical approach.

A good agreement was obtained (Fig. 5) between the curves of the directly measured racquet-head velocity and the racquet-head velocity determined using the method of component summation [equation (14)]. It is apparent that the residual between the two curves is in the form of a small oscillating 'noise' component whose effects are more pronounced early on in the forward swing during periods of low racquet-head speed.

DISCUSSION

The algorithm developed in this paper was designed to examine the combined effect that the instantaneous angular velocities of the racquet-arm segments and the position of the racquet-head have on racquet-head speed. It is recognized that such an approach does not lead to an understanding of how the joint torques are created during the swing. However, given the magnitude of the errors associated with calculated joint torques (Williams and Cavanagh, 1983), one questions how much understanding would be gained from such an approach. The main advantages of the angular velocity/racquet-head position approach are: (a) the computation errors are markedly reduced since only the first derivative of the displacement data is required; (b) the position of the racquet-head during a segment's rotation is taken into account when evaluating a segment's rotational contribution to racquet-head speed; (c) joint angular velocities are easier than accelerations for the coach/athlete to assimilate when linking the results with the film or video images; and (d) generally, once a player is made aware of a rotational deficiency in either timing or magnitude, the necessary corrections to the swing pattern are best realized by examining the corresponding rotational technique employed by highly skilled players.

For the tennis player that we examined in this study, there was no evidence of forearm supination occurring either just prior or just after ball impact, as had previously been reported by Van Gheluwe *et al.*

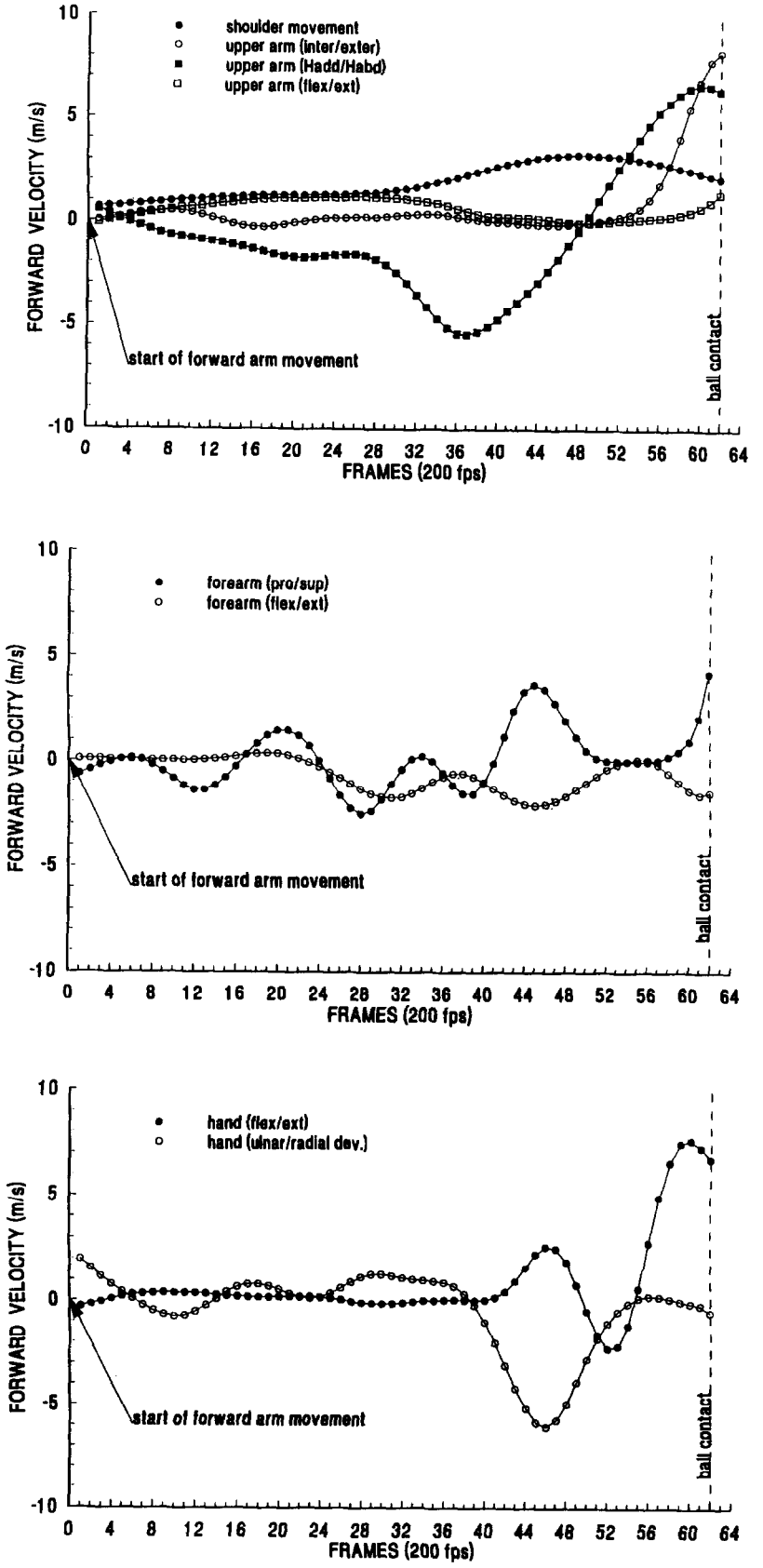

Fig. 4. Individual contributions to the forward speed of the racquet-head during a tennis serve provided by the rotations of the upper arm and shoulder (top), forearm (middle), and hand (bottom).

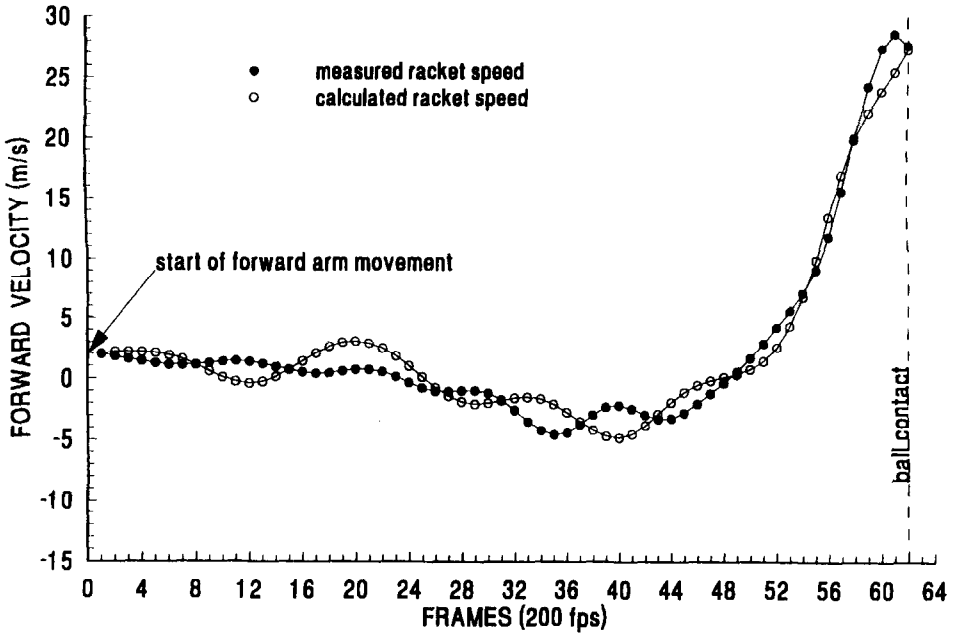

Fig. 5. A comparison of the summated contributions of the upper limb segments (as well as that of the legs and torso which are indirectly estimated from the shoulder velocity), with the instantaneous measured racquet-head speed for the tennis serve.

(1987). The measured racquet-head speed of 27 $m\,s^{-1}$ for the center of the racquet was below the mean value of 31.1 $m\,s^{-1}$ reported by Elliott *et al.* (1986) for high-performance male players. This result is consistent with the observation that this player's serve lacked speed.

It is apparent from the results that one cannot judge the effectiveness of a segment's rotation in producing racquet-head velocity unless both the magnitude of the rotation and the position of the racquet-head with respect to the segment's rotational axis are known. It must be kept in mind that while a segment's rotation in itself may provide little or no direct contribution to racquet-head speed, the rotation may position the racquet-head further from another segment's active rotational axis, thus indirectly contributing to racquet-head speed. Also, the possibility of a proximal segment's rotation facilitating a greater muscle torque at a more distal joint later in the swing sequence must be taken into consideration when proposing any alterations to the player's serve mechanics.

While the overall results of the component summation agreed well with the measured racquet-head speed in the forward direction (Fig. 5), there is certainly noise in the component summation curve. Examining the individual graphs, it is evident that the main source of this noise originates from the inaccuracy of the measurements for pronation/supination of the forearm (Fig. 4—middle). Similar measurement noise was reported by Woo and Chapman (1991) for the forearm pronation movement in squash. The inaccuracies in the calculations of the forearm's longitudinal rotation are directly attributable to the small magnitude of the displacement vector $\mathbf{r}_{B/C}$. Such a small displacement vector requires accurate determination of the digitized points B and A, an accuracy that is generally not possible when using 16 mm film and only three cameras. Future efforts should attempt to extend these markers further from the longitudinal axis of the forearm.

In summary, this procedure provides a quantitative means of assessing the effectiveness of a player's arm segment rotations in producing racquet-head speed. While the tennis serve was used as the example in this paper, the method could just as easily have been applied to the swing mechanics of any of the racquet sports. Different rotational techniques employed by different players can be evaluated for their effectiveness in producing racquet-head speed. Such a record of rotational techniques would provide insight into the alterations that might possibly be made in the swing mechanics of the developing player.

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